ADRIAN MATIAS

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León, Spain

PROFILE

Data scientist and ML engineer with 11 years of experience taking machine learning from problem framing to production. In adtech: install-prediction models behind mobile user-acquisition bidding (Appodeal), ad ranking and Bayesian bandits on the DSP side (Chartboost), persona modelling and attribution for programmatic ads (Digilant). Also fraud detection, NLP (RavenPack), recommender systems and large-scale experimentation on Spark and Databricks. Most recent GenAI work: an LLM-as-judge product that scores software vendors from their raw documentation, with cited evidence. Owned the typed output contract and the correctness gates, and built the human ground-truth lane that checks the judge against expert scores.

EXPERIENCE

Senior AI Engineer

Cavendish Professionals → PwC Contractor cavendish-professionals | León, Spain – Remote | *May 2026 – Sep 2026*

Senior engineer on an LLM-as-judge product that scores software vendors. Built the human ground-truth lane alone: domain-expert template parser, vendor matching, band placement, tier-weighted averaging and roll-up. It reconciles human evaluator scores with model scores, so the judge's accuracy can be checked against ground truth.

KEY ACHIEVEMENTS

- Contributed to every component of the product architecture in a team of three senior AI engineers, a PM and the associate product owner.
- Introduced and owned schema-driven, domain-driven and spec-driven development: business requirements kept as a contract artifact that agentic development builds against.
- Owned the typed output contract and the correctness gate set (type baselines per file and rule, a complexity ratchet, mutation tests, required status checks), and designed and managed the end-to-end validation workflows behind them.
- Created and owned full epics end to end, from business requirement to shipped increment.
- Ran agentic coding on the production codebase as a measured practice: cost and turns, tuned per model tier.

Data Scientist → DS Technical Lead

Appodeal: appodeal.com | León, Spain – Remote | *Dec 2024 – Jan 2026*

Developed install prediction models for mobile user acquisition: cold start, transformers, user embeddings, transfer learning, MLOps, production pipelines, parity, experimentation and monitoring. Promoted to DS technical lead in December 2025.

KEY ACHIEVEMENTS

- Replaced separate per-advertiser and per-platform models with one attention model per target, with segment identity as a feature. Launched on 5% of bid requests; later it covered all Android CPI, then iOS.
- Designed and deployed cold start for advertiser onboarding in two ways: transfer learning through an affinity tower trained on a wider set of interactions, and cold-advertiser masking validated on held-out cold entities. New advertisers bid from day one from the shared model, with eCPI better than curated whitelists for inventory and demand segments. Custom PyTorch.
- Resolved parity mismatches and calibration gaps: traced discrepancies in internal transformations until the models received identical inputs offline and online and their outputs matched through the inference server; made parity a workflow with unit tests in Databricks.
- Proposed and designed the user embedding system, built from users' interactions with apps; delivered with transfer learning at performance parity and orders-of-magnitude lower cost.

Senior Data Scientist

Quant AI Lab: quant.global | León, Spain – Remote | *Jul 2024 – Dec 2024*

Designed and delivered end-to-end ML solutions for international clients; reviewed technical scope, aligned with stakeholders, and supervised model productivization.

KEY CONTRIBUTIONS

- Led the turnaround of an underperforming client ML system for utility fraud detection and ranking: found the architectural faults, fixed them, and restored accuracy, efficiency and client satisfaction.
- Found critical faults in the data-ingestion and feature-engineering pipelines.
- Led development of a real-time credit card fraud classifier combining scalable classical ML with a deep VAE. Open version: github.com/adrianmatias/ml-sandbox/tree/master/tx_class

Senior Data Scientist

Chartboost: chartboost.com | León, Spain – Remote | *Jun 2023 – Jun 2024*

Developed ad ranking and reinforcement learning systems for in-app game advertising; improved DSP performance through large-scale experiments and behavioral analysis.

KEY ACHIEVEMENTS

- Evolved and monitored a Bayesian-bandit RL system with policy-evaluation tools; designed online/offline evaluation frameworks for it.
- Built a two-tower neural ranking model for ads.
- Ran post-mortems on adversarial DSP dynamics and model feedback loops.

Machine Learning Engineer

RavenPack | León, Spain – Remote | *Mar 2022 – Jan 2023*

Researched, implemented, and maintained text analytics, NLU, and semantic search products on massive financial news data; productionized prototypes with emphasis on MLOps and reproducibility.

KEY ACHIEVEMENTS

- Few-shot search personalization: simplified architecture, reduced error 30%.
- Active learning PoC for learn-to-rank: projected 300% annotation efficiency gain.
- Long-text fingerprinting algorithm for duplicate content detection.

EARLIER ROLES

Numbrs · *Machine Learning Engineer Aug 2020 – Oct 2021*
Extended Deep Learning / Active Learning for transaction processing; scaled classifier across countries with GDPR-compliant pipelines.

Stratio BD · *Data Scientist Aug 2019 – Aug 2020*
Propensity/churn on Spark; BERT+graph email segmentation; multi-source weather DB (Elasticsearch/Flask).

Digilant · *Machine Learning Engineer Jun 2015 – Apr 2017*
Redesigned Consumer Persona ML model; Game Theory attribution prototype; co-directed Master's thesis.

IEBS School · *Teacher Oct 2019 – Oct 2021*
Webinars and masterclasses on real-time Big Data, AI, and ML applications.

DOCOMO DIGITAL · *Data Scientist Apr 2017 – Aug 2019*
Led fraud detection (99.8% specificity, <50 ms) and default-payment filtering; mentored juniors.

2007–2015 · *Tunnel Site Engineer & Maths & Sciences Tutor*
Civil/tunnel engineering; university teaching: Math, Physics, Chemistry.

FREELANCE

Freelance Data Scientist / ML Engineer | *Mar 2022 – Jun 2025*

On-demand projects in semantic search, time series forecasting, and portfolio optimization for financial analysis.

SKILLS

Core Technologies

Python, FastAPI, Pydantic, PostgreSQL, PyTorch, Transformers, Azure OpenAI / Azure / AKS, AWS/S3, Databricks, PySpark, Pandas, SQL, Docker, MLflow, Git, GitHub Actions, AI-augmented tools (Ollama, llama.cpp, Hermes Agent)

Machine Learning & GenAI

LLMs, RAG, agentic systems, LangChain / LangGraph-style orchestration, document extraction & provenance, scoring engines, Deep Learning (attention, embeddings, transfer learning), Recommender Systems, NLP, RL / Bayesian Bandits, MLOps (parity, A/B, monitoring), Few-Shot / Active Learning, adtech ranking and install prediction for bidding (DSP, mobile UA)

Engineering Practices

Clean code, typed contracts as source of truth, CI drift gates, OOP & FP, TDD / local e2e harnesses, reproducible pipelines, Agile / XP, production ownership

Consulting & Soft Skills

Multi-client consulting, stakeholder communication, mentoring, correctness gatekeeping, simplification

SELECTED COURSES & CERTIFICATIONS

- Claude Code: A Highly Agentic Coding Assistant: Coursera (2026)
- ML Engineering for Production (MLOps) Specialization: Coursera (2021)
- Deep Learning with PyTorch: Siamese Networks (2022)
- Deep Learning Specialization: Coursera (2016)
- Structuring Machine Learning Projects: Coursera (2018)
- Apache Spark for Big Data & Scalable ML: Coursera (9.9/10, 2015)
- Functional Programming in Scala Specialization: Coursera (9.2/10, 2017)
- Bitcoin and Cryptocurrency Technologies: Coursera (9.5/10, 2017)
- Algorithms & Data Structures with Java (I, II): Coursera (8.5/10, 2014)
- Analytics Edge: Data Science & Predictive Models: MITx (9.3/10, 2014)
- Introduction to Programming & Data Science Using Python (I, II): MITx (8.7/10, 2014)

SELECTED PROJECTS

- **opo — AI-compiled exam preparation**: opo.taoaieng.io Live paid product for Spanish public-sector exams: official law and past papers compiled once into a cited corpus of 28 topic pages and 660 tagged official items, then a 40-item diagnostic allocates the student's hours by official weight × mastery deficit. Six proven CI gates block release; no claim ships without a citation.
- **not the tao with adrian matias**: youtube.com/@notTheTaoWithAdrianMatias Podcast exploring simplicity, consciousness, reality, machine learning, coding, sovereign AI.
- **RAG Blog System**: github.com/adrianmatias/ml-sandbox/tree/master/ragblog RAG over a personal blog archive with local open models: web crawling, vector embeddings and LLMs, evaluated with Ragas using separate generator and judge models.
- **The Mirror**: https://github.com/adrianmatias/the-mirror Knowledge repository built on llm-wiki for a personal AI assistant: notes and synthesis that persist across sessions.
- **Credit Card Transaction Fraud Classifier**: github.com/adrianmatias/ml-sandbox/tree/master/tx_class Real-time fraud detection API (classical + unsupervised signals). Deploy-ready: FastAPI, Docker, RayServe.

EDUCATION

- **Master's in Geological Engineering**: University of Oviedo (2010–2012)
- **Degree in Mining & Energy Engineering**: University of León (2012)
- **Bachelor in Mining Engineering**: University of León (2003–2007)

References available on request